

Analysis 1

Mock Mid-semester Examination

Time: 3 hours

Marks: 100

- Answer *any four* questions. Each question carries 25 marks.
- You may use that $\mathbb{R}$ satisfies the l.u.b. axiom. You may also use the definitions of the limit of a sequence, limsup and liminf of a sequence, subsequence of a sequence, convergence of a series, and continuity and uniform continuity of a function.

I. Let $\langle a_n : n \geq 1 \rangle$ be a bounded sequence. For $n \geq 1$ let $E_n = \{a_k : k \geq n\}$ and let $t_n = \text{l. u. b. } E_n$.

- (a) Show that there is $t \in \mathbb{R}$ such that $\lim_{n \rightarrow \infty} t_n = t$.
- (b) Show that there is a subsequence $\langle b_k : k \geq 1 \rangle$ of $\langle a_n : n \geq 1 \rangle$ converging to t .
- (c) Suppose that $\langle a_n : n \geq 1 \rangle$ is Cauchy. Show that it converges to t .

II. Let $\langle a_n : n \geq 1 \rangle \subseteq (0, \infty)$ be a sequence and for $n \geq 1$ let $b_n = \max\{a_j : j \leq n\}$ and $c_n = a_n/b_n$.

- (a) Suppose $\lim_n c_n = \theta < 1$. Show that $\lim_n a_n = \alpha < \infty$.
- (b) Suppose $\limsup_n c_n = \theta < 1$. Show that $\limsup_n a_n = \alpha < \infty$.
- (c) Suppose $a_n \neq a_m$ whenever $n \neq m$, and $\lim_n c_n = 1$. Give examples to show that $\limsup_n a_n$ could be finite or infinite.

III. Suppose the series $\sum_k a_k$ and $\sum_k b_k$ are convergent. Let $c_k = a_k b_k$ for $k \geq 1$.

- (a) Give an example to show that $\sum_k c_k$ may not converge.
- (b) If in addition $a_k \geq 0$ and $b_k \geq 0$ for all k , show that $\sum_k c_k$ converges.

IV. Let $A \subseteq \mathbb{R}$ and let f, g be bounded continuous functions from A to $\mathbb{R}$.

- (a) Suppose $g(x) \neq 0$ for all $x \in A$. For $x \in A$ let $h(x) = f(x)/g(x)$. Show that h is continuous.
- (b) Suppose $A = [0, 1]$. Show that f is uniformly continuous.
- (c) Suppose $A = (0, 1)$. Give an example to show that f need not be uniformly continuous.

V. Let $\langle a_n : n \geq 1 \rangle$ be such that $\limsup_n \sqrt[n]{|a_n|} = 2$. For $x \in \mathbb{R}$ let $f_n(x) = \sum_{k=1}^n a_k x^k$.

- (a) Show that $f_n(x)$ converges for $|x| < \frac{1}{2}$.
- (b) Show that $f_n(x)$ does not converge for $|x| > \frac{1}{2}$.
- (c) For $|x| < \frac{1}{2}$ let $f(x) = \lim_n f_n(x)$. Show that f is continuous on $(-\frac{1}{2}, \frac{1}{2})$.